Class X Maths (Standard) Sample Paper – Set 2

Max. Marks: 80 **Time Allowed: 3 hours**

General Instructions:

1. This Question Paper has 5 Sections A, B, C, D and E.

2. In **Section A**, Questions no. 1-18 are multiple choice questions (MCQs) and questions no.

19 and 20 are Assertion- Reason based questions of 1 mark each.

3. **Section B** has 5 questions carrying 02 marks each.

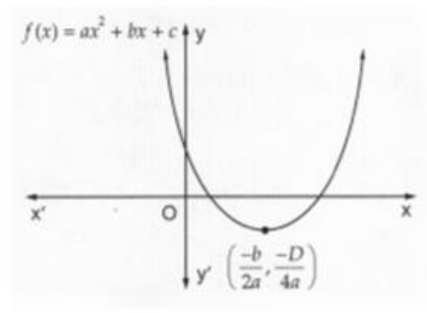
4. **Section C** has 6 questions carrying 03 marks each.

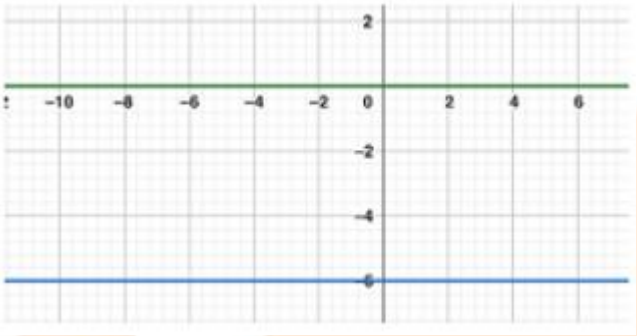
5. **Section D** has 4 questions carrying 05 marks each. Maximum Marks: 80

6. **Section E** has 3 case based integrated units of assessment (04 marks each) with sub- parts of the values of 1, 1 and 2 marks each respectively.

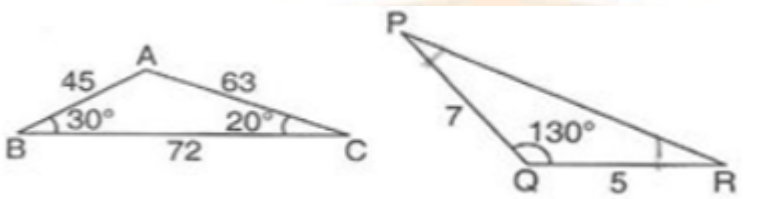
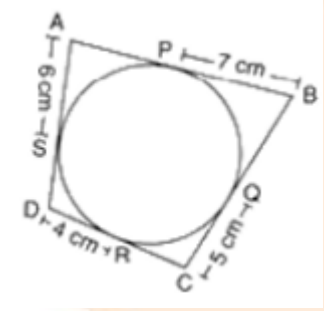
7. All Questions are compulsory. However, an internal choice in 2 Qs of 5 marks, 2 Qs of 3 marks and 2 Questions of 2 marks has been provided. An internal choice has been provided in the 2marks questions of Section E.

8. Draw neat figures wherever required. Take $\pi = \frac{22}{7}$ wherever required if not stated.

Section A		20
Q. No.	**Questions**	**Marks**
1.	$7 \times 11 \times 13 + 13$ is a/an: a) odd number but not composite b) square number c) prime number d) composite number	1
2.	Figure show the graph of the polynomial $f(x) = ax^2 + bx + c$ for which  a) $a > 0$, $b < 0$ and $c > 0$ b) $a < 0$, $b < 0$ and $c < 0$ c) $a < 0$, $b > 0$ and $c > 0$ d) $a > 0$, $b > 0$ and $c < 0$	1

3.	The pair of linear equations y = 0 and y = - 6 has:  a) no solution b) only solution (0, 0) c) infinitely many solutions d) a unique solution	1
4.	In the Maths Olympiad of 2020 at Animal Planet, two representatives from the donkey's side, while solving a quadratic equation, committed the following mistakes. i. One of them made a mistake in the constant term and got the roots as 5 and 9. ii. Another one committed an error in the coefficient of x and he got the roots as 12 and 4. But in the meantime, they realised that they are wrong and they managed to get it right jointly. Find the quadratic equation. a) $2x^2 + 7x - 24 = 0$ b) $x^2 + 4x + 14 = 0$	1

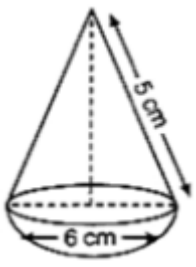
	c) $3x^2 - 17x + 52 = 0$ d) $x^2 - 14x + 48 = 0$	
5.	If S_n denote the sum of n terms of an A.P. with first term a and common difference d such that is independent of x, then a) a = 2d b) d = a c) d = - a d) d = 2a	1
6.	ABCD is a rectangle whose three vertices are B (4,0), C (4,3) and D (0,3). The length of one of its diagonals is a) 5 b) 3 c) 4 d) 2	1
7.	The coordinates of the mid-point of the line segment joining the points (-2, 3) and (4, -5) are a) (0, 0) b) (-1, 1) c) (1, -1)	1

	d) (-2, 4	
8.	In the figures find the measures of $\angle P$ and $\angle R$  a) $20^{\circ}, 30^{\circ}$ b) $50^{\circ}, 40^{\circ}$ c) $30^{\circ}, 20^{\circ}$ d) $40^{\circ}, 50^{\circ}$	1
9.	In the given figure, the perimeter of ABCD is  a) 44 cm b) 36 cm c) 40 cm d) 48 cm	1

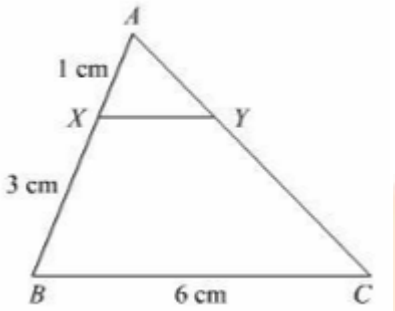
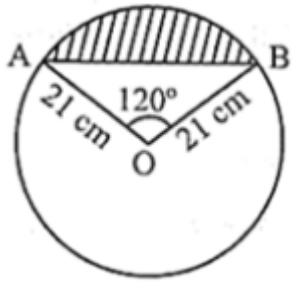
10.	A circle inscribed in △ABC having AB = 10 cm, BC = 12 cm, CA = 28 cm touching sides at D, E, F (respectively). Then AD + BE + CF = _____.  a) 22 cm b) 25 cm c) 18 cm d) 20 cm	1
11.	If $\sin\theta + \cos\theta = \sqrt{2}\cos\theta$, then the value of $\cos\theta - \sin\theta$ is a) $\sqrt{2}\sin\theta$ b) $3\sin\theta$ c) $\sin\theta$ d) $2\sin\theta$	1
12.	If $\cos\theta = \frac{4}{5}$ then $\tan\ \theta = ?$ a) $\frac{3}{4}$	1

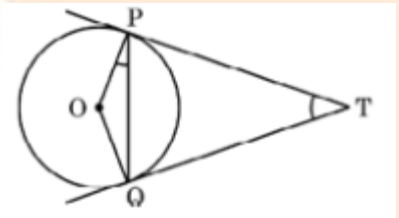
	b) $\frac{5}{3}$ c) $\frac{4}{3}$ d) $\frac{3}{5}$	
13.	An observer 1.5 m tall is 28.5 m away from a tower and the angle of elevation of the top of the tower from the eye of the observer is 45°. The height of the tower is a) 30 m b) 26.5 m c) 28.5 m d) 27	1
14.	In a circle of radius 14 cm, an arc subtends an angle of 120° at the centre. If $\sqrt{3} = 1.73$ then the area of the segment of the circle is a) 124.63 cm^2 c) 120.56 cm^2 b) 130.57 cm^2 d) 118.24 cm^2	1
15.	Area of a segment of a circle of radius r and central angle 90° is:	1

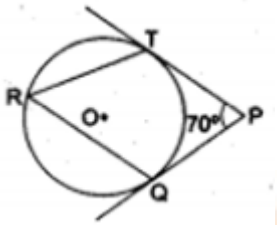
	a) $\frac{2\pi r}{4}-\frac{1}{2}r^2$ b) $\frac{\pi r^2}{4}-\frac{1}{2}r^2$ c) $\frac{\pi^2}{2}-\frac{1}{2}r^2$ d) $\frac{2\pi r}{4}-r^2\sin 90^\circ$	
16.	A die is thrown once. The probability of getting an even number is a) $\frac{1}{3}$ b) $\frac{5}{6}$ c) $\frac{1}{6}$ d) $\frac{1}{2}$	1
17.	A bag contains 5 red balls and n green balls. If the probability of drawing a green ball is three times that of a red ball, then the value of n is: a) 20 b) 18 c) 15	1

	d) 10	
18.	The distribution below gives the marks obtained by 80 students on a test: Marks: Less than 10 | Less than 20 | Less than 30 | Less than 40 | Less than 50 | Less than 60 Number of Students: 3 | 12 | 27 | 57 | 75 | 80 The modal class of this distribution is: a) 30 - 40 b) 20 - 30 c) 50 - 60 d) 10 - 2	1
19.	**Assertion (A):** The given figure represents a hemisphere surmounted by a conical block of wood. The diameter of their bases is 6 cm each and the slant height of the cone is 5 cm. The volume of the solid is 196 cm^3  	1

	Reason (R): The volume hemisphere is given by $\frac{2}{3}\pi r^3$ a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A. c) A is true but R is false. d) A is false but R is true	
20.	**Assertion (A):** Sum of first n terms in an A.P. is given by the formula: $S_n = 2n\,[2a + (n - 1)d]$ **Reason (R):** Sum of first 15 terms of 2 , 5 , 8 ... is 345. a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A. c) A is true but R is false. d) A is false but R is true	1
	Section B	**10**
21.	Find the least number which when divided by 12, 16 and 24 leaves remainder 7 in each case.	2
22.	In the given figure XY‖BC. Find the length of XY.	2

		
23.	If a circle touches the side BC of a triangle ABC at P and extended sides AB and AC at Q and R, respectively, prove that AQ = $\frac{1}{2}$ (BC+CA+AB)	2
24.	Show that: $\tan^4\theta + \tan^2\theta = \sec^4\theta - \sec^2\theta$ for $0^\circ \le \theta \ge 90^\circ$ **OR** Prove that: $\frac{\tan\theta}{1-\cot\theta} + \frac{\cot\theta}{1-\tan\theta} = 1 + \tan\theta + \cot\theta$	2
25.	Find the area of the segment shown in Fig., if radius of the circle is 21 cm and $\angle AOB = 120^0$ (Use $\pi = \frac{22}{7}$)  **OR**	2

	A car has two wipers which do not overlap. Each wiper has a blade of length 25cm sweeping through an angle of 115°. Find the total area cleaned at each sweep of the blades.	
	Section C	**18**
26.	Prove that $3 + \sqrt{5}$ is an irrational number.	3
27.	f α and β are zeroes of the quadratic polynomial $4x^2 + 4x + 1$, then form a quadratic polynomial whose zeroes are 2 α and 2 β.	3
28.	A two-digit number is 4 times the sum of its digits. If 18 is added to the number, the digits are reversed. Find the number. **OR** A plane left 30 minutes late than its scheduled time and in order to reach the destination 1500 km away in time, it had to increase its speed by 100 km/h from the usual speed. Find its usual speed.	3
29.	Two tangents TP and TQ are drawn to a circle with centre O from an external point T. Prove that $\angle PTQ = 2\ \angle OPQ$.  **OR** In the given figure, O is the centre of a circle. PT and PQ are tangents to the circle from an external point P. If $\angle TPQ = 70°$, find $\angle TRQ$.	3



30. Prove that: $\frac{\tan\theta+\sec\theta-1}{\tan\theta-\sec\theta+1}=\frac{1+\sin\theta}{\cos\theta}$ — 3

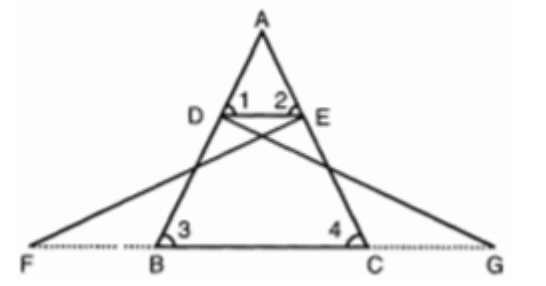
31. Find median for the following data: — 3

Class Interval	**Frequency**
10-19	2
20-29	4
30-39	8
40-49	9
50-59	4
60-69	2
70-79	1

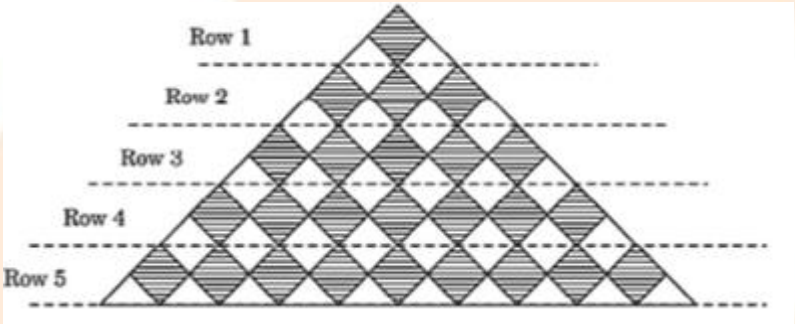
Section D — **20**

32. Solve: $\frac{x-1}{2x+1}+\frac{2x+1}{x-1}=2, x\neq-\frac{1}{2},1$ — 5

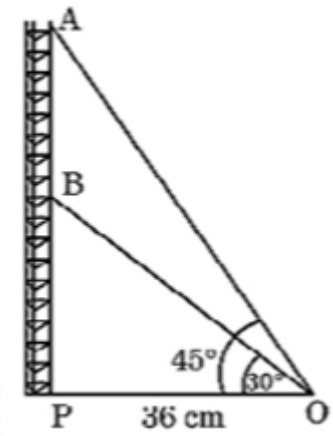
OR

	₹ 9000 were divided equally among a certain number of persons. Had there been 20 more persons, each would have got ₹ 160 less. Find the original number of persons.	
33.	In the following figure, and △FEC ≅△GBD ∠1 = ∠2 Prove that △ADE≅△ABC.  	5
34.	A solid is composed of a cylinder with hemispherical ends. If the whole length of the solid is 104 cm and the radius of each of the hemispherical ends is 7 cm, find the cost of polishing its surface at the rate of ₹10 per dm^2. **OR** A solid is in the form of a cylinder with hemispherical ends. The total height of the solid is 19 cm and the diameter of the cylinder is 7 cm. Find the volume and total surface area of the solid (Use = 22/7)	5
35.	Find the missing frequencies in the following distribution, if the sum of the frequencies is 120 and the mean is 50.	5

Class	0-20	20-40	40-60	60-80	80-100
Frequency	17	f_1	32	f_2	19

	Section E	**12**
36	**Read the text carefully and answer the questions:** A fashion designer is designing a fabric pattern. In each row, there are some shaded squares and unshaded triangles.  i) Identify A.P. for the number of squares in each row. ii) Identify A.P. for the number of triangles in each row. **OR** Write a formula for finding total number of triangles in n number of rows. Hence, find S_{10}. iii) If each shaded square is of side 2 cm, then find the shaded area when 15 rows have been designed.	4
37.	**Read the text carefully and answer the questions:** In order to facilitate smooth passage of the parade, movement of traffic on certain roads leading to the route of the Parade and Tableaux ah rays restricted. To avoid traffic on the road Delhi Police decided to construct a rectangular route plan, as shown in the figure.	4

	 i) If Q is the mid point of BC, then what are the coordinates of Q? ii) What is the length of the sides of quadrilateral PQRS? **OR** What is the length of route ABCD? iii) What is the length of route PQRS?	
38.	**Read the text carefully and answer the questions:** Radio towers are used for transmitting a range of communication services including radio and television. The tower will either act as an antenna itself or support one or more antennas on its structure. On a similar concept, a radio station tower was built in two Sections A and B. Tower is supported by wires from a point O. Distance between the base of the tower and point O is 36 cm. From point O, the angle of elevation of the top of the Section B is 30o and the angle of elevation of the top of Section A is 45^0.	4



i) Find the length of the wire from the point O to the top of Section B.

ii) Find the distance AB.

OR

Find the height of the Section A from the base of the tower.

iii) Find the area of ΔOPB.

Answer Key

Section A

1. d) composite number

Explanation: We have $7 \times 11 \times 13 + 13 = 13\ (77 + 1) = 13 \times 78$ itself, therefore it is a composite number.

2. a) $a > 0$, $b < 0$ and $c > 0$

Explanation: Clearly, $f(x) = ax^2 + bx + c$ represent a parabola opening upwards. Therefore, $a > 0$.

The vertex of the parabola is in the fourth quadrant, therefore $b < 0$.

$y = ax^2 + bx + c$ cuts the Y-axis at P which lies on OY.

Putting $x = 0$ in $y = ax^2 + bx + c$, we get $y = c$.

So the coordinates of P is $(0, c)$.

Clearly, P lies on $OY \Rightarrow c > 0$.

Hence, $a > 0$, $b < 0$ and $c > 0$.

3. a) no solution

Explanation: Since, we have y = 0 and y = -6 are two parallel lines.

Therefore, no solution exists.

4. d) $x^2 - 14x + 48 = 0$

Explanation: For 1st one,

Let the equation be $x^2 + ax + b = 0$

Since roots are 5 and 9

$\therefore a = -14$ and $b = 45$

For 2nd one,

Let the equation be $x^2 + px + q = 0$

Since roots are 12 and 4.

$\therefore p = -16$ and $q = 48$

Now, according to the question, b and p both are wrong.

Therefore, the correct equation would be $x^2 - 14x + 48 = 0$

5. d) $d = 2a$

Explanation: Given AP in which

First term $= a$

Common difference $= d$

Number of terms $= n$

Given S_n denotes the sum of n terms

So, $S_{kx} = \frac{kx}{2}[2a + (kx-1)d]$

and $S_x = \frac{x}{2}[2a + (x-1)d]$

Now,

$$\frac{S_x}{S_{kx}} = \frac{\left(\frac{x}{2}\right)[2a + (x-1)d]}{\left(\frac{kx}{2}\right)[2a + (kx-1)d]}$$

$$= \frac{x[2a + (x-1)d]}{k[2a + (kx-1)d]}$$

$$= \frac{k[2a + (x-1)d]}{[2a + xd - d]}$$

$$= \frac{k[2a + kx - d]}{k[(2a - d) + kxd]}$$

If $d = 2a$, then

$$\frac{S_x}{S_{kx}} = \frac{[(2a - 2a) + x \times 2a]}{[kx\{(2a - 2a) + kx \times 2a\}]}$$

$$= \frac{x \times 2a}{k^2 \times x \times 2a}$$

$$= \frac{1}{k^2}$$

6. a) 5

Explanation: Three vertices of a rectangle $ABCD$ are $B(4,0)$, $C(4,3)$, and $D(0,3)$.

Length of one of its diagonals:

$$BD = \sqrt{(4-0)^2 + (0-3)^2} = \sqrt{4^2 + 3^2}$$

$$= \sqrt{16+9} = \sqrt{25} = 5$$

7. c) (1, -1)

Explanation: Let the coordinates of midpoint $C(x, y)$ of the line segment joining the points $A(-2,3)$ and $B(4,-5)$.

$$\therefore x = \frac{x_1 + x_2}{2} = \frac{-2+4}{2} = \frac{2}{2} = 1$$

And

$$y = \frac{y_1 + y_2}{2} = \frac{3+(-5)}{2} = \frac{-2}{2} = -1$$

Therefore, the coordinates of mid-point C are $(1,-1)$.

8. a) $20^\circ, 30^\circ$.

Explanation: In triangle ABC, $\angle A + \angle B + \angle C = 180^\circ$

$$\Rightarrow \angle A + 30^\circ + 20^\circ = 180^\circ$$

$\Rightarrow \angle A = 130^{\circ}$

In triangle \$ABC\$ and QRP, $\frac{AB}{QR} = \frac{AC}{PQ}$

$\Rightarrow \frac{45}{5} = \frac{63}{7} \Rightarrow \frac{9}{1} = \frac{9}{1}$

Since sides of triangles ABC and QRP are proportional, and included angles are equal, therefore by SAS similarity criteria,

$\Delta ABC \sim \Delta QRP$

$\angle A = \angle Q, \angle B = \angle R, \angle C = \angle P$

$\Rightarrow \angle P = 20^{\circ}, \angle R = 30^{\circ}$

9. a) 44 cm

Explanation: Since tangents from an external point to a circle are equal in length.

$\therefore$ AS = AP = 6 cm and AB = 6 + 7 = 13 cm

PB = BQ = 7 cm and BC = 7 + 5 = 12 cm

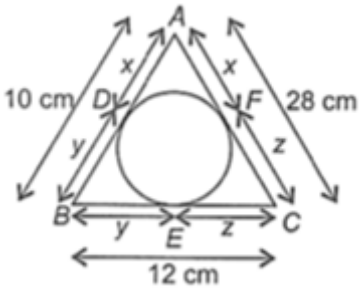
CQ = CR = 5 cm and CD = 5 + 4 = 9 cm

RD = SD = 4 cm and AD = 4 + 6 = 10 cm

Therefore, perimeter of quadrilateral ABCD = 13 + 12 + 9 + 10 = 44 cm

10. b) 25 cm

Explanation:



x + y = 10 cm ...(i)

y + z = 12 cm ...(ii)

x + z = 28 cm ...(iii)

Adding (i), (ii) and (iii), we get

2(x + y + z) = 50

x + y + z = 25 cm

11. a) $\sqrt{2}\sin\theta$

Explanation: Given: $\sin\theta + \cos\theta = \sqrt{2}\cos\theta$

Squaring both sides, we get

$\Rightarrow \sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta = 2\cos^2\theta$

$\Rightarrow \cos^2\theta - 2\sin\theta\cos\theta = \sin^2\theta$

$\Rightarrow \cos^2\theta - 2\sin\theta\cos\theta + \sin^2\theta = 2\sin^2\theta$

$\Rightarrow (\cos\theta - \sin\theta)^2 = 2\sin^2\theta$

$\Rightarrow \cos\theta - \sin\theta = \sqrt{2}\sin\theta$

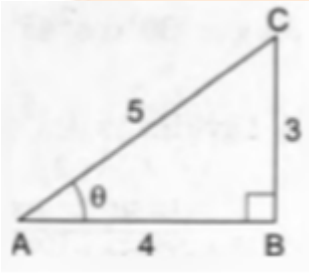
12. a) $\frac{3}{4}$

Explanation: $\cos\theta = \frac{4}{5} = \frac{AB}{AC}$

$\therefore BC^2 = AC^2 - AB^2 = 25 - 16 = 9$

$\Rightarrow BC = 3$

$\therefore \tan\theta = \frac{BC}{AB} = \frac{3}{4}$



13. a) 30 m

Explanation: Let AB be the observer and $CD = h$ metres be the tower.

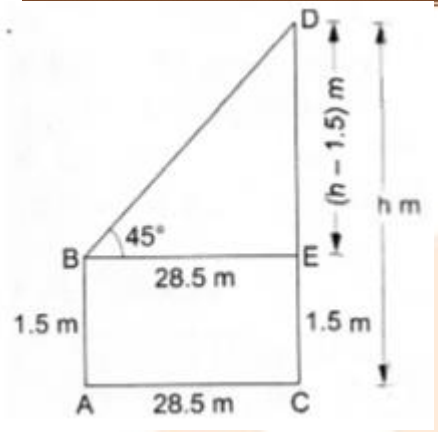
$BE = AC = 28.5\text{m}$.

From right $\triangle BED$, we have

$\frac{DE}{BE} = \tan 45^\circ \Rightarrow \frac{DE}{28.5} = 1$

$\Rightarrow DE = 28.5\text{m}$

$\therefore h - 1.5 = 28.5 \Rightarrow h = 30.$



14. c) $120.56\ \text{cm}^2$

Explanation: $\text{ar(segment)} = \left(\frac{\pi r^2\theta}{360} - r^2 \sin\frac{\theta}{2}\cos\frac{\theta}{2}\right)$

$= \left(\frac{22}{7}\times 14\times 14\times\frac{120}{360}\right) - \left(14\times 14\times \sin 60^\circ \cos 60^\circ\right)$

$= \left(\frac{616}{3} - \frac{\sqrt{3}}{2}\times\frac{1}{2}\times 14\times 14\right)\text{cm}^2$

$= (205.33 - 49\times 1.73)\text{cm}^2$

$= (205.33 - 84.77)\text{cm}^2$

$= 120.56\text{cm}^2$

15. b) $\frac{\pi r^2}{4} - \frac{1}{2}r^2$

16. d) $\frac{1}{2}$

Explanation: Number of all possible outcomes = 6.

Even numbers are 2,4, 6. Their number is 3.

$\therefore$ P (getting an even number) = $\frac{3}{6} = \frac{1}{2}$

17. c) 15

Explanation: Given, Number of red ball $= 5$

Number of green ball $= n$

$\therefore$ Total ball $= n + 5$

Now P(red ball) $= \frac{5}{n+5}$

and P(green ball) $= \frac{n}{n+5}$

Now, according to the question

$\frac{n}{n+5} = \frac{3 \times 5}{n+5}$

$n = 15$

So, number of green ball $= 15$

18. a) 30 – 40

19. d) A is false but R is true.

20. a) Both A and R are true and R is the correct explanation of A.

Section B

21. LCM of 12, 16, 24 = 48

Required number is 48 + 7 = 55.

22. Given $XY \parallel BC$

$AX = 1\text{cm}$, $XB = 3\text{cm}$, and $BC = 6\text{cm}$

$AB = AX + XB$

$= 1 + 3 = 4\text{cm}$

In ΔAXY and ΔABC:

$\angle A = \angle A[\text{Common}]$

$\angle AXY = \angle ABC[\text{Correspondingangles}]$

Then, $\triangle AXY \sim \triangle ABC$ [By AA similarity]

$\therefore \frac{AX}{AB} = \frac{XY}{BC}[\text{Correspondingpartsofsimilar\$}\triangle\text{\$areproportional}]$

$\Rightarrow \frac{1}{4} = \frac{XY}{6}$

$\Rightarrow XY = \frac{6}{4} = 1.5cm$

23. $AQ = \frac{1}{2}(2AQ)$

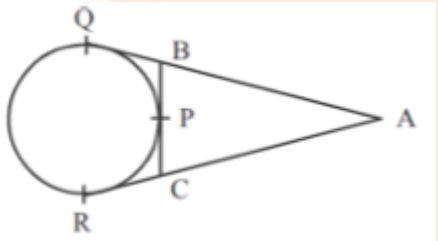
$= \frac{1}{2}(AQ + AQ)$

$= \frac{1}{2}(AQ + AR)$

$= \frac{1}{2}(AB + BQ + AC + CR)$

$= \frac{1}{2}(AB + BC + CA)$

$\therefore [BQ = BP, CR = CP]$



24. We have,

$\text{LHS} = \sin^4 A - \cos^4 A$

$\Rightarrow \text{LHS} = (\sin^2 A)^2 - (\cos^2 A)^2$

$\Rightarrow \text{LHS} = (\sin^2 A + \cos^2 A)(\sin^2 A - \cos^2 A)$

$\Rightarrow \text{LHS} = \sin^2 A - \cos^2 A \left[\because \sin^2 A + \cos^2 A = 1\right]$

$\Rightarrow \text{LHS} = \sin^2 A - (1 - \sin^2 A) = 2\sin^2 A - 1$

$\Rightarrow \text{LHS} = 2(1 - \cos^2 A) - 1 = 1 - 2\cos^2 A = \text{RHS}$

OR

$\text{LHS} = \dfrac{1 - \cos A}{1 + \cos A}$

Multiplying numerator and denominator by 1 - cos A:

$= \dfrac{(1 - \cos A)(1 - \cos A)}{(1 + \cos A)(1 - \cos A)}$

$= \dfrac{(1 - \cos A)^2}{1 - \cos^2 A} \quad \left[\because (a + b)(a - b) = a^2 - b^2\right]$

$= \dfrac{(1 - \cos A)^2}{\sin^2 A} \quad \left[\because 1 - \cos^2 A = \sin^2 A\right]$

$= \left(\dfrac{1 - \cos A}{\sin A}\right)^2$

$= \left(\dfrac{1}{\sin A} - \dfrac{\cos A}{\sin A}\right)^2$

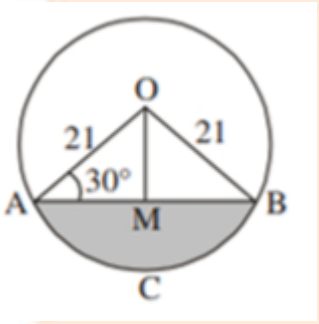
$= (\csc A - \cot A)^2 \quad \left[\because \frac{1}{\sin A} = \csc A, \frac{\cos A}{\sin A} = \cot A\right]$

$= [-1(\cot A - \csc A)]^2$

$= (\cot A - \csc A)^2$

Hence proved.

25. Draw OM $\perp$ AB



$\angle OAB = \angle OBA = 30^\circ$

$\sin 30^\circ = \frac{1}{2} = \frac{OM}{21} \Rightarrow OM = \frac{21}{2}$

$\cos 30^\circ = \frac{\sqrt{3}}{2} = \frac{AM}{21} \Rightarrow AM = \frac{21}{2}\sqrt{3}$

Area of $\triangle OAB$:

$= \frac{1}{2} \times AB \times OM = \frac{1}{2} \times 21\sqrt{3} \times \frac{21}{2}$

$= \frac{441}{4}\sqrt{3}\,cm^2$

$\therefore$ Area of shaded region = Area (sector OACB) – Area($\triangle OAB$)

$$= \frac{22}{7} \times 21 \times 21 \times \frac{120}{360} - \frac{441}{4}\sqrt{3}$$

$$= \left(462 - \frac{441\sqrt{3}}{4}\right) \text{cm}^2 \text{ or } 271.3 \text{cm}^2 \text{ (approx.)}$$

OR

Radius of each wiper $= 25\text{cm}$, Angle $= 115^\circ$

$\therefore \theta = 115^\circ$

Total area cleaned at each sweep of the blades:

$$= 2\left[\frac{115}{360} \times \frac{22}{7} \times 25 \times 25\right] \quad \left(\because \text{Area} = \frac{\theta}{360}\pi r^2\right)$$

$$= 2\left[\frac{230 \times 22 \times 5 \times 25}{72 \times 7}\right]$$

$$= 2\left[\frac{230 \times 11 \times 125}{36 \times 7}\right]$$

$$= 2\left[\frac{115 \times 11 \times 125}{18 \times 7}\right]$$

$$= 2\left[\frac{158125}{126}\right] \text{cm}^2$$

$$= 1254.96 \text{cm}^2$$

Section C

26. Let $3+\sqrt{5}$ is a rational number.

$3+\sqrt{5}=\frac{p}{q}, q \neq 0$

$3+\sqrt{5}=\frac{p}{q}$

$\Rightarrow \sqrt{5}=\frac{p}{q}-3$

$\Rightarrow \sqrt{5}=\frac{p-3q}{q}$

Now, in RHS $\frac{p-3q}{q}$ is rational.

This shows that $\sqrt{5}$ is rational.

But this contradicts the fact that $\sqrt{5}$ is irrational. This is because we assumed that $3+\sqrt{5}$ is a rational number.

$\therefore 3+\sqrt{5}$ is an irrational number.

27. Let the given polynomial is $p(x)=4x^2+4x+1$

Since, α, β are zeroes of $p(x)$,

$\therefore \alpha+\beta=$ sum of zeroes $=\frac{-4}{4}=-1$

Also, $\alpha \cdot \beta$ = Product of zeroes $= \frac{1}{4}$

Now a quadratic polynomial whose zeroes are 2α and 2β:

$x^2 - (\text{sum of zeroes})x + \text{Product of zeroes}$

$= x^2 - (2\alpha + 2\beta)x + 2\alpha \times 2\beta$

$= x^2 - 2(\alpha + \beta)x + 4(\alpha\beta)$

$= x^2 - 2(-1)x + 4 \times \frac{1}{4}$

$= x^2 + 2x + 1$

The quadratic polynomial whose zeroes are 2α and 2β is $x^2 + 2x + 1$.

28. Let us suppose that the digit at unit place be x

Suppose the digit at tens place be y.

Thus, the number is 10y + x.

According to question it is given that the number is 4 times the sum of the two digits.

Therefore, we have

10y + x = 4(x + y)

⇒ 10y + x = 4x + 4y

⇒ 4x + 4y - 10y - x = 0

⇒ 3x - 6y = 0

$\Rightarrow 3(x - 2y) = 0$

$\Rightarrow x - 2y = 0$

After interchanging the digits, the number becomes $10x + y$.

Again according to question If 18 is added to the number, the digits are reversed.

Thus, we have $(10y + x) + 18 = 10x + y$

$\Rightarrow 10x + y - 10y - x = 18$

$\Rightarrow 9x - 9y = 18$

$\Rightarrow x - y = \frac{18}{9}$

$\Rightarrow x - y = 2$

Therefore, we have the following systems of equations

$x - 2y = 0$ .....(1)

$x - y = 2$.....(2)

Here x and y are unknowns. Now let us solve the above systems of equations for x and y.

Subtracting the equation (1) from the (2), we get

$(x - y) - (x - 2y) = 2 - 0$

$\Rightarrow x - y - x + 2y = 2$

$\Rightarrow y = 2$

Now, substitute the value of y in equation (1), we get

$x - 2 \times 2\Rightarrow = 0$

$\Rightarrow x - 4 = 0$

$\Rightarrow x = 4$

Therefore the number is $10 \times 2 + 4 = 24$

Thus the number is 24.

OR

Let the usual speed of the plane = x km/hr.

Distance to the destination = 1500 km

Case (i):

We know that, Speed $= \frac{\text{Distance}}{\text{Time}}$

$\Rightarrow \text{Time} = \frac{\text{Distance}}{\text{Speed}}$

So, in case (i) Time $= \frac{1500}{x}$ Hrs

Case (ii):

Distance to the destination $= 1500\text{km}$

Increased speed $= 100\text{km/hr}$

So, speed $= x + 100$

So, in case (ii) Time $= \frac{1500}{x+100}$ Hrs

So, according to the question:

$\therefore \frac{1500}{x} - \frac{1500}{x+100} = \frac{30}{60}$

$\Rightarrow x^2 + 100x - 300000 = 0$

$\Rightarrow x^2 + 600x - 500x - 300000 = 0$

$\Rightarrow (x+600)(x-500) = 0$

$\Rightarrow x = 500 \text{or} x = -600$

Since, speed cannot be negative, x = 500.

Therefore, Speed of plane = 500 km/hr.

29. Given : A circle with centre O and an external point T and two tangents TP and TQ to the circle, where P, Q are the points of contact.

To Prove: $\angle PTQ = 2\angle OPQ$

Proof:

Let $\angle PTQ = \theta$

Since TP, TQ are tangents drawn from point T to the circle.

TP = TQ

$\therefore$ TPQ is an isoscles triangle

$\therefore \angle TPQ = \angle TQP = \frac{1}{2}(180^0 - \theta) = 90^0 - \frac{\theta}{2}$

Since, TP is a tangent to the circle at point of contact P

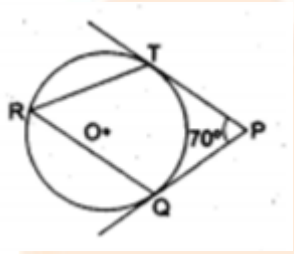
$\therefore \angle OPT = 90^\circ$

$\therefore \angle \text{OPQ} = \angle \text{OPT} - \angle \text{TPQ} = 90^\circ - (90^\circ - \frac{1}{2}\theta) = \frac{\theta}{2} = \frac{1}{2}\angle PTQ$

Thus, $\angle \text{PTQ} = 2\angle \text{OPQ}$

OR

In the given figure, O is the centre of a circle. PT and PQ are tangents to the circle from an external point P. If $\angle \text{TPQ} = 70^0$ then,we have to find $\angle \text{TRQ}$.



We know that the radius and tangent are perpendicular at their point of contact.

$\angle OTP = \angle OQP = 90^\circ$

Now, in quadrilateral OQPT, applying the angle sum property of a quadrilateral:

$\angle OQT + \angle OTP + \angle OQP + \angle OTPQ = 360^\circ$

Substitute the known values:

$\angle OQT + 90^\circ + 90^\circ + 70^\circ = 360^\circ$

Simplify:

$\angle OQT + 250^\circ = 360^\circ$

Solve for $\angle OQT$:

$\angle OQT = 360^\circ - 250^\circ = 110^\circ$

Next, we use the property that the angle subtended by an arc at the center is double the angle subtended by the arc at any point on the circumference of the circle.

Therefore, we can write:

$$\angle TRQ = \frac{1}{2}\angle OQT$$

Substitute the value of $\angle OQT$:

$$\angle TRQ = \frac{1}{2}\times 110^{\circ} = 55^{\circ}$$

30. We have,

$$LHS = \frac{\tan\theta + \sec\theta - 1}{\tan\theta - \sec\theta + 1}$$

$$\Rightarrow LHS = \frac{(\tan\theta + \sec\theta) - 1}{(\tan\theta - \sec\theta) + 1}$$

$$\Rightarrow LHS = \frac{(\sec\theta + \tan\theta) - (\sec^2\theta - \tan^2\theta)}{\tan\theta - \sec\theta + 1}$$

$$\left[\text{Since } \sec^2\theta - \tan^2\theta = 1\right]$$

$$\Rightarrow LHS = \frac{(\sec\theta + \tan\theta) - (\sec\theta + \tan\theta)(\sec\theta - \theta)}{\tan\theta - \sec\theta + 1}$$

$$\Rightarrow LHS = \frac{(\sec\theta + \tan\theta)\left[1 - (\sec\theta - \tan\theta)\right]}{\tan\theta - \sec\theta + 1}$$

$$\Rightarrow LHS = \frac{\sec\theta + \tan\theta}{(\tan\theta - \sec\theta + 1)}$$

$$\Rightarrow LHS = \sec\theta + \tan\theta = \frac{1}{\cos\theta} + \frac{\sin\theta}{\cos\theta} = \frac{1+\sin\theta}{\cos\theta}$$

$$\Rightarrow LHS = \text{RHS}$$

31. Calculation of Median:

Class Interval	**Frequency**	**c.f.**
9.5-19.5	2	2
19.5-29.5	4	6
29.5-39.5	8	14
39.5-49.5	9	23
49.5-59.5	4	27
59.5-69.5	2	29
69.5-79.5	1	30

$n = 30, \quad \frac{n}{2} = 15, \quad \text{Median class} = 39.5 - 49.5$

$l = 39.5, \quad c = 14, \quad f = 9, \quad h = 10$

The formula for Median:

$$\text{Median} = l + \left(\frac{\frac{n}{2} - f}{h} \right) \times 10$$

Substitute the values:

$$\text{Median} = 39.5 + \left(\frac{15-14}{9}\right)\times 10$$

$$= 39.5 + \frac{1}{9}\times 10$$

$$= 39.5 + 1.11 = 40.61$$

Section D

32. Given:

$$\frac{x-1}{2x+1} + \frac{2x+1}{x-1} = 0$$

Let $\frac{x-1}{2x+1} = y$, so

$$\frac{2x+1}{x-1} = \frac{1}{y}$$

Substituting this value:

$$y + \frac{1}{y} = 2 \quad \text{or} \quad y^2 + 1 = 2y$$

$$\Rightarrow y^2 - 2y + 1 = 0 \quad \text{or} \quad (y-1)^2 = 0$$

Putting $y = \frac{x-1}{2x+1}$:

$$\frac{x-1}{2x+1} = 1 \quad \text{or} \quad x - 1 = 2x + 1$$

or $\Rightarrow x = -2$

OR

Let the original number of persons be x.

Total amount to be divided among all people = Rs. 9000/-

So, the share of each person is: $\frac{9000}{x}$

If the number of persons is increased by 20, then the new share of each person is: $\frac{9000}{x+20}$

According to the question:

$$\frac{9000}{x}-\frac{9000}{x+20}=160$$

$$\Rightarrow \frac{9000}{x(x+20)}=160$$

$$\frac{9000}{x}+\frac{180000}{x(x+20)}=160$$

$$\Rightarrow \frac{180000}{x^2+20x}=160$$

$$\Rightarrow \frac{180000}{160}=x^2+20x$$

$$\Rightarrow 1125=x^2+20x$$

$$\Rightarrow x^2+20x-1125=0$$

$$\Rightarrow x^2+45x-25x-1125=0$$

$$\Rightarrow x(x+45)-25(x+45)=0$$

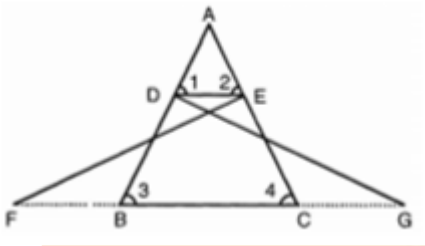
$\Rightarrow (x+45)(x-25)=0$

$\Rightarrow x-25=0$ [The number of persons cannot be negative, thus $x+45\neq 0$]

$\Rightarrow x=25$

Hence, the original number of persons is 25.

33.



$\therefore \Delta FEC \cong \Delta GBD$

$\Rightarrow EC = BD$ (i)

It is given that $\angle 1 = \angle 2$, so

$\Rightarrow AE = AD$ (Isosceles triangle property) (ii)

From equations (i) and (ii),

$\frac{AE}{EC} = \frac{AD}{DB}$

$\Rightarrow DE \parallel BC$ (converse of B.P.T.)

$\Rightarrow \angle 1 = \angle 3$ and $\angle 2 = \angle 4$ (Corresponding angles)

Thus in ΔADE and ΔABC

$\angle A = \angle A$

$\angle 1 = \angle 3$

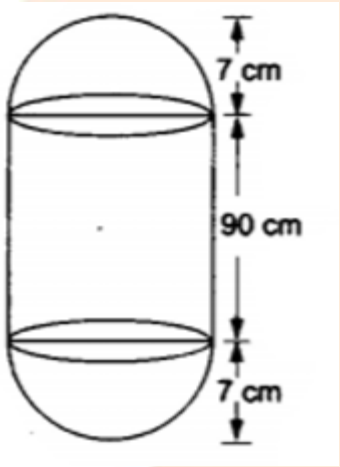
$\angle 2 = \angle 4$

$\Delta ADE \sim \Delta ABC$ (AAA criterion of similarity)

$\Delta ADE \sim \Delta ABC$

Hence proved.

34.



Radius of each hemispherical end = 7 cm.

Height of each hemispherical part = its radius = 7 cm.

Height of the cylindrical part = (104 - 2 + × 7) cm = 90 cm.

Area of surface to be polished = 2(curved surface area of the hemisphere) + (curved surface area of the cylinder)

Surface area $= 2\left(2\pi r^2 + 2\pi rh\right)$ sq units

$$= \left[\left(4 \times \frac{22}{7} \times 7 \times 7\right) + \left(2 \times \frac{22}{7} \times 7 \times 90\right)\right] \text{ cm}^2$$

$$= (616 + 3960) \text{ cm}^2 = 4576 \text{ cm}^2$$

$$= \frac{4576}{10 \times 10} \text{ dm}^2 = 45.76 \text{ dm}^2 \quad [\text{since } 10\text{cm} = 1\text{dm}]$$

Therefore, Cost of polishing the surface of the solid = ₹ $(45.76 \times 10) = 457.60$

OR

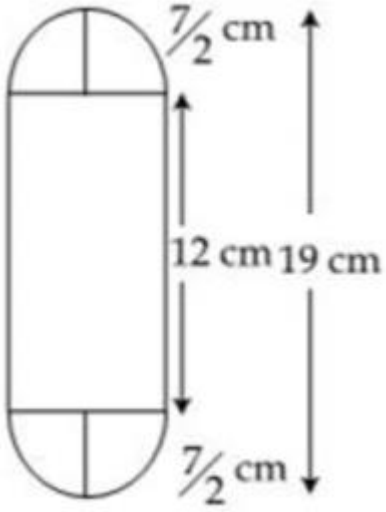
Diameter of the cylinder = 7 cm

Therefore radius of the cylinder = $\frac{7}{2}$ cm

Total height of the solid = 19 cm

Therefore, Height of the cylinder portion = 19 - 7 = 12 cm

Also, radius of hemisphere = $\frac{7}{2}$ cm



Let V be the volume and S be the surface area of the solid. Then,

V = Volume of the cylinder + Volume of two hemispheres

$$\Rightarrow V = \left\{ \pi r^2 h + 2\left(\frac{2}{3}\pi r^3 \right) \right\} \text{cm}^3$$

$$\Rightarrow V = \pi r^2 \left(h + \frac{4r}{3} \right) \text{cm}^3$$

$$\Rightarrow V = \left(\frac{22}{7} \times \left(\frac{7}{2} \right)^2 \right) \times \left(12 + \frac{4}{3} \times \frac{7}{2} \right) \text{cm}^3$$

$$= \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times \left(12 + \frac{4}{3} \times \frac{7}{2} \right) = 641.66 \text{cm}^3$$

And,

S = Curved surface area of cylinder + Surface area of two hemispheres}

$\Rightarrow S = \left(2\pi rh + 2 \times 2\pi r^2\right)\text{cm}^2$

$\Rightarrow S = 2\pi r(h + 2r)\text{cm}^2$

$\Rightarrow S = 2 \times \frac{22}{7} \times \frac{7}{2} \times \left(12 + 2 \times \frac{7}{2}\right)\text{cm}^2$

$= \left(2 \times \frac{22}{7} \times \frac{7}{2}\right) \times \left(12 + 2 \times \frac{7}{2}\right)\text{cm}^2 = 418\text{cm}^2$

35.

Class Interval	**Frequency** f_i	**Mid-Value** x_i	$f_i x_i$
0-20	17	10	170
20-40	f_1	30	$30\ f_1$
40-60	32	50	1600
60-80	f_2	70	$70\ f_2$
80-100	19	90	1710
	$\sum f_i = 68 + f_1 + f_2 = 120$		$\sum f_i x_i = 3480 + 30 f_1 + 70 f_2$

Given mean = 50, we have:

$$\text{Mean} = \frac{\sum f_i x_i}{\sum f_i}$$

$\Rightarrow 50 = \frac{3480 + 30 f_1 + 70 f_2}{120}$

$\Rightarrow 6000 = 3480 + 30 f_1 + 70 f_2$

$\Rightarrow 30 f_1 + 70 f_2 = 252$ ... (i)

Also, $68 + f_1 + f_2 = 120$, so

$f_1 = 52 - f_2$

Substituting this into equation (i), we get:

$3(52 - f_2) + 7 f_2 = 252$

$\Rightarrow 4 f_2 = 96$

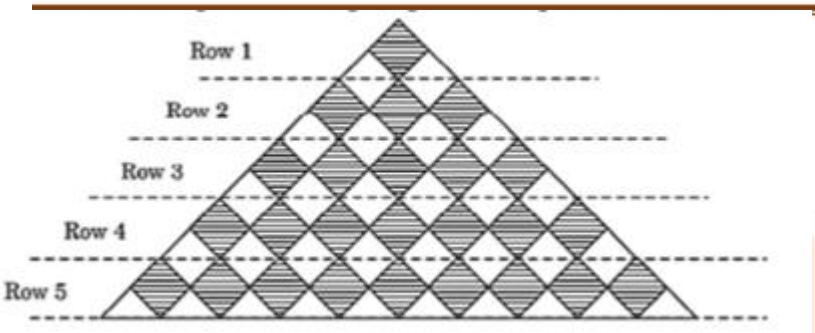
$\Rightarrow f_2 = 24$

$\Rightarrow f_1 = 52 - 24 = 28$

Hence, $f_1 = 28$ and $f_2 = 24$.

Section E

36. A fashion designer is designing a fabric pattern. In each row, there are some shaded squares and unshaded triangles.



(i) A.P. for the number of squares in each row is 1, 3, 5, 7, 9 …

(ii) A.P. for the number of triangles in each row is 2, 6, 10, 14 ...

OR

$S_n = \frac{n}{2}\left[4+(n-1)4\right] = 2n^2$

$\therefore S_{10} = 2\times 10^2 = 200$

(iii) Area of each square = $2\times 2 = 4\ cm^2$

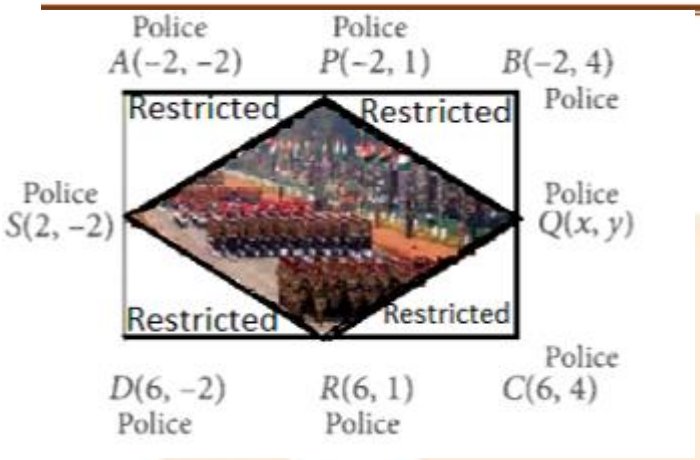
Number of squares in 15 rows:

$\frac{15}{2}\left(2+14\times 2\right) = 225$

Shaded area:

$225\times 4 = 900\text{cm}^2$

37. In order to facilitate smooth passage of the parade, movement of traffic on certain roads leading to the route of the Parade and Tableaux ah rays restricted. To avoid traffic on the road Delhi Police decided to construct a rectangular route plan, as shown in the figure.



(i) Q(x, y) is the mid-point of B(-2, 4) and C(6, 4) , so

$$\therefore (x, y) = \left(\frac{-2+6}{2}, \frac{4+4}{2}\right) = (2, 4)$$

(ii) Since PQRS is a rhombus, therefore PQ = QR = RS = PS .

$$\therefore PQ = \sqrt{(-2-2)^2 + (1-4)^2} = \sqrt{16+9} = \sqrt{25} = 5 \text{ units}$$

Thus, the length of each side of PQRS is 5 units.

OR

Length of CD = 4 + 2 = 6 units and length of AD = 6 + 2 = 8 units.

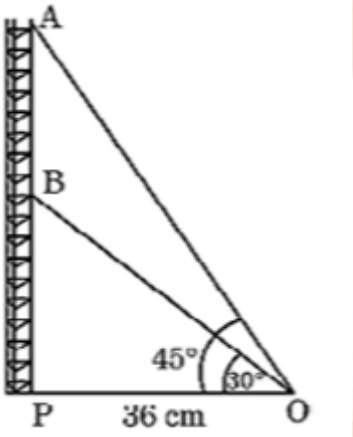
$\therefore$ Length of route $ABCD = 2(6+8) = 28$ units

(iii) Length of route PQRS = $4 \times PQ = 4 \times 5 = 20$ units

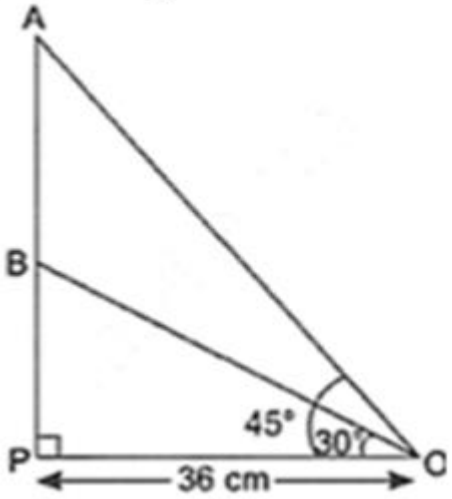
38. Radio towers are used for transmitting a range of communication services including radio and television. The tower will either act as an antenna itself or support one or more antennas on its structure. On a similar concept, a radio station tower was built in two Sections A and B.

Tower is supported by wires from a point O.

Distance between the base of the tower and point O is 36 cm. From point O, the angle of elevation of the top of the Section B is 30^{0} and the angle of elevation of the top of Section A is 45^{0}



(i) Let the length of wire BO = x cm



$$\therefore \cos 30^{\circ} = \frac{PO}{BO}$$

$$\Rightarrow \frac{\sqrt{3}}{2} = \frac{36}{x}$$

$\Rightarrow x = \frac{36 \times 2}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$

$= 12 \times \frac{2\sqrt{3}}{3}$

$= 24\sqrt{3}\text{cm}$

(ii) In ΔAPO, $\tan 45^\circ = \frac{AP}{PO}$

$\Rightarrow 1 = \frac{AP}{36}$

$\Rightarrow AP = 36\text{cm}\ldots(\text{i})$

Now, In ΔPBO,

$\tan 30^\circ = \frac{BP}{PO}$

$\Rightarrow \frac{1}{\sqrt{3}} = \frac{BP}{36}$

$\Rightarrow BP = \frac{36}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$

$= \frac{36\sqrt{3}}{3}$

$= 12\sqrt{3}\text{cm}$

$\therefore AB = AP - BP$

$= 36 - 12\sqrt{3}\text{cm}$

OR

In ΔAPO, $\tan 45^\circ = \frac{AP}{36}$

$1 = \frac{AP}{36}$

$A = 36\text{cm}$

Height of section A from the base of the tower $= AP = 36\text{cm}$.

(iii) In ΔOPB,

$\tan 30^\circ = \frac{BP}{PO}$

$\Rightarrow \frac{1}{\sqrt{3}} = \frac{BP}{36}$

$BP = \frac{36}{\sqrt{3}}$

$= \frac{36}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$

$= 12\sqrt{3}\text{cm}$

Now, Area of $\Delta OPB = \frac{1}{2} \times \text{height} \times \text{base}$

$= \frac{1}{2} \times BP \times OP$

$= \frac{1}{2} \times 12\sqrt{3} \times 36$

$= 216\sqrt{3}\text{cm}^2$